Expectation

RV takes a real value each time the random experiment is performed. Expectation seeks to represent 'average' of these values over a large # of experiments.

Motivation - X discrete RV. pmf P_X, support $\mathcal{E}_X$

Say N experiments. N_k - # times $X = k$

We expect $\frac{N_k}{N} \to P_X(k)$ { we will formalize this later in the course }

We take $\boxed{E(X) = \lim_{N\to\infty} \sum_{k \in \mathcal{E}_X} k \cdot \frac{N_k}{N}} \to \boxed{\sum_{k \in \mathcal{E}_X} k P_X(k)}$

$\mathcal{E}_X$: value; $\frac{N_k}{N}$: fraction of times k obs.

Def - X discrete RV with pmf P_X

$$E[X] \triangleq \sum_{k \in \mathcal{E}_X} k \cdot P_X(k)$$

[weighted avg of values X takes.]

Def - X cont RV $\quad E[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx$

[general defn of expectation ?]

Eg

① $U \sim \text{Unif}[a, b]$ $\quad E[U] = \int_a^b \frac{1}{b-a} x\, dx$

$$= \frac{a+b}{2}$$

(2) $X \sim Exp(\mu)$ $\quad E[X] = \int_0^\infty x \cdot \mu e^{-\mu x} dx$

$= \mu \int_0^\infty x e^{-\mu x} dx \overset{t=\mu x}{=} \frac{1}{\mu} \underbrace{\int_0^\infty y e^{-y} dy}_{\text{Erlang dist } (\lambda=1)} = \frac{1}{\mu}.$

(3) $N \sim Bin(n, p)$

$E[N] = \sum_{k=0}^{n} k \cdot \binom{n}{k} p^k (1-p)^{n-k}$

$= (1-p)^n \sum_{k=0}^{n} k \binom{n}{k} \left(\frac{p}{1-p}\right)^k$

$(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k$. Diff. both sides

$n(1+x)^{n-1} = \sum_{k=0}^{n} k \binom{n}{k} x^{k-1} \Rightarrow nx(1+x)^{n-1} = \sum_{k=0}^{n} k \binom{n}{k} x^k$

So $E[N] = (1-p)^n \cdot n \left(\frac{p}{1-p}\right) \left(1 + \frac{p}{1-p}\right)^{n-1}$

$= np$

(4) $X \sim N(\mu, \sigma^2)$

$E[X] = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} x \cdot e^{-(x-\mu)^2/2\sigma^2} dx$

$= \mu + \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} (x-\mu) e^{-(x-\mu)^2/2\sigma^2} dx$

$= \mu + \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} y e^{-y^2/2\sigma^2} dy \quad [y = x - \mu]$

$$= \mu + \frac{1}{\sqrt{2\pi\sigma^2}} \left[\int_0^\infty y e^{-y^2/2\sigma^2} dy + \int_{-\infty}^0 y e^{-y^2/2\sigma^2} dy \right]$$

$$\int_0^\infty \frac{y}{\sigma^2} e^{-y^2/2\sigma^2} dy = \sigma^2 \qquad -\sigma^2$$

Rayleigh dist.

So $E[X] = \mu$.

Show - $N \sim Geo(p)$ $\quad P_X(k) = (1-p)^{k-1} p \quad k \geq 1$

$$E[X] = \frac{1}{p}$$

$N \sim Poi(\lambda)$ - $E[X] = \lambda$

(5) $X \sim$ Cauchy RV $\quad f_X(x) = \frac{1}{\pi(1+x^2)} \quad x \in \mathbb{R}$

$$E[X] = \int_{-\infty}^\infty \frac{x}{\pi(1+x^2)} dx = \int_0^\infty \frac{x}{\pi(1+x^2)} dx + \int_{-\infty}^0 \frac{x\,dx}{\pi(1+x^2)}$$

$t = 1+x^2$

$$= \frac{1}{2\pi} \int_1^\infty \frac{dt}{t} \qquad = -\infty$$

$$= \frac{1}{2\pi} [\ln t]_1^\infty$$

$$= \infty$$

Expectation $E[X]$ undefined

(6) $X \sim \text{Pareto}(x_0, \alpha) \qquad x_0, \alpha > 0$

$$\mathbb{P}(X > x) = \begin{cases} \left(\frac{x}{x_0}\right)^{-\alpha} & x \geq x_0 \\ 1 & x < x_0 \end{cases}$$

$$f_X(x) = \begin{cases} \frac{\alpha}{x_0^{-\alpha}} \cdot x^{-(\alpha+1)} & x \geq x_0 \\ 0 & 0 \cdot \omega. \end{cases}$$

$$\mathbb{E}[X] = \int_{x_0}^{\infty} \frac{\alpha}{x_0^{-\alpha}} x \cdot x^{-(\alpha+1)} \, dx = \int_{x_0}^{\infty} \frac{\alpha}{x_0^{-\alpha}} x^{-\alpha} \, dx$$

$$= \frac{\alpha}{x_0^{-\alpha}} \left[\frac{x^{1-\alpha}}{1-\alpha} \right]_{x_0}^{\infty} \qquad \text{for } \alpha \neq 1$$

$$\Rightarrow \mathbb{E}[X] = \begin{cases} \left(\frac{\alpha}{\alpha-1}\right) x_0 & \alpha > 1 \\ \infty & \alpha \leq 1 \end{cases}$$

Infinite expectation for $\alpha \leq 1$. This is because tail probabilities don't decay exponentially, only polynomially.

Moments of RV

(1) For positive integer k, k^{th} moment of RV X is defined as $\mathbb{E}[X^k]$

(2) Variance of RV X is defined as $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}(X))^2]$

Captures "spread" of distribution around the mean.

$\sqrt{\text{Var}(X)} \rightarrow$ Std. deviation

Clearly $\text{Var}(X) \geq 0$

Lemma – $\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}^2(X)$

Pf $\mathbb{E}\left((X - \mathbb{E}(X))^2\right) \triangleq \text{Var}(X)$. Say X is discrete. $\mu \triangleq \mathbb{E}(X)$

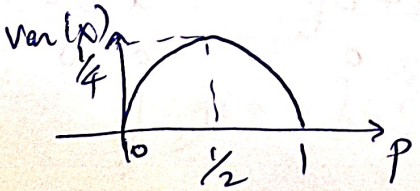
Then $\text{Var}(X) = \sum_{x \in \mathcal{E}_X} p_X(x) \cdot (x - \mu)^2$

$$= \sum_{x \in \mathcal{E}_X} p_X(x) \cdot (x^2 + \mu^2 - 2\mu x)$$

$$= \sum p_X(x) \cdot x^2 + \sum p_X(x) \cdot \mu^2 - 2\mu \cdot \underbrace{\sum p_X(x) \cdot x}_{\mu}$$

$$= \mathbb{E}[X^2] + \mu^2 - 2\mu^2 = \mathbb{E}(X^2) - \mathbb{E}^2(X)$$

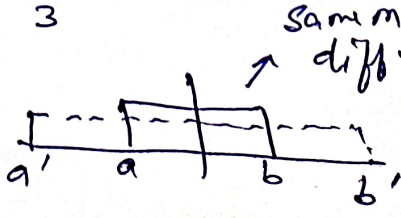
Eg $X \sim \text{Ber}(p)$, $\mathbb{E}[X] = p$. $\mathbb{E}(X^2) = p$. $\text{Var}(X) = p - p^2 = p(1-p)$



Eg $X \sim \text{Unif}(a, b)$ $\mathbb{E}(X) = \frac{a+b}{2}$

$$\mathbb{E}(X^2) = \int_0^1 x^2 \cdot \frac{1}{b-a}\, dx = \frac{b^3 - a^3}{3(b-a)} = \frac{b^2 + ab + a^2}{3}$$

$$\text{Var}(X) = \mathbb{E}(X^2) - \mathbb{E}^2(X) = \frac{(b-a)^2}{12}$$



Eg - $X \sim N(\mu, \sigma^2)$ $\quad \text{Var}(X) = \sigma^2$ Prove!

For Random vector $\bar{X}$, for $g(\bar{X})$

$$E[g(\bar{X})] = \begin{cases} \sum_{x \in \mathcal{E}_{\bar{X}}} g(\bar{x}) \cdot p_{\bar{X}}(\bar{x}) & \text{discrete} \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} g(\bar{x}) \cdot f_{\bar{X}}(\bar{x})\, d\bar{x} & \text{cont.} \end{cases}$$

For $g(X,Y)$

$$E[g(X,Y)] = \begin{cases} \sum_{x,y} g(x,y)\, p_{XY}(x,y) \\ \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y)\, f_{XY}(x,y)\, dx\, dy \end{cases}$$

Properties of Expectation

(1) Linearity (a) $E[aX] = a\,E[X]$, (b) $E[X+Y] = E[X] + E[Y]$

Pf (a) → easy, substitute in formula

(b) $E[X+Y] = \sum_x \sum_y (x+y)\, p_{XY}(x,y)$ ~~$= \sum_y \sum_x x\, p_{XY}(x,y) + \sum_y p_{XY}(x,y)$~~

Say X, Y discrete

$$= \sum_x \sum_y x\, p_{XY}(x,y) + \sum_x \sum_y y \cdot p_{XY}(x,y)$$

$$= \sum_x x \cdot p_X(x) + \sum_y y \cdot p_Y(y) = E[X] + E[Y]$$

Note - Always true, doesn't need $X \perp\!\!\!\perp Y$.

Each ball dropped in bin chosen at random.

Eg m balls, n bins. Find $\mathbb{E}[\#$ of empty bins$]$

Define $X_i = \begin{cases} 1 & \text{if bin } i \text{ empty} \\ 0 & \text{o.w} \end{cases}$ Indicator RV

$S = \sum_{i=1}^{n} X_i \rightarrow \#$ of empty bins.

$\mathbb{E}[X_i] = \mathbb{P}(\text{bin } i \text{ is empty}) = \left(\frac{n-1}{n}\right)^m$

So $\mathbb{E}[S] = \mathbb{E}\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} \mathbb{E}[X_i] = n \cdot \left(\frac{n-1}{n}\right)^m$

Are X_i, X_j ind ?

(2) Independence – If X, Y ind. $\Rightarrow \mathbb{E}[XY] = \mathbb{E}[X] \cdot \mathbb{E}[Y]$

Pf – say X, Y discrete. $\mathbb{E}[XY] = \sum_x \sum_y xy \, P_{XY}(x,y)$

$= \sum_{x,y} xy \, P_X(x) \cdot P_Y(y) = \left(\sum_x x P_X(x)\right)\left(\sum_y y P_Y(y)\right)$

$= \mathbb{E}[X] \cdot \mathbb{E}[Y]$

If $\mathbb{E}[XY] = \mathbb{E}[X] \cdot \mathbb{E}[Y]$, then X, Y are said to be uncorrelated. $\mathbb{E}[XY]$ is called correlation b/w X, Y.

$\mathbb{E}[(X - \mu_X)(Y - \mu_Y)] \rightarrow$ covariance b/w X, Y ($\text{cov}(X,Y)$)

$= \mathbb{E}[XY] - \mu_X \mu_Y$

So if X, Y uncorrelated $\Leftrightarrow \text{cov}(X,Y) = 0$

If $\mathbb{E}[XY] = 0 \Rightarrow X, Y$ said to be orthogonal

If we have a random vector $X = (X_1, X_2 \dots X_n)$

Covariance matrix $(n \times n)$ $\begin{bmatrix} M_{ij} \end{bmatrix}$ $M_{ij} = \text{cov}(X_i, X_j)$

Diagonal elements are $\{\text{Var}(X_i)\}_{i=1}^{n}$

If X, Y are independent $\Rightarrow$ X, Y are uncorrelated

But the opposite is not necessarily true!

Eg $X \in \{-2, -1, 1, 2\}$ w.p. $\frac{1}{4}$ each

$Y \in \{-1, 1\}$ w.p. $\frac{1}{2}$ each

X, Y ind.

Say $Z = XY$. Are Z, X ind? Are Z, X uncorrelated?

$$\mathbb{E}[ZX] = \mathbb{E}[X^2 Y] = \mathbb{E}[X^2] \cdot \mathbb{E}[Y] = 0$$

Now $\mathbb{E}[X] = 0$ so $\mathbb{E}[ZX] = \mathbb{E}[Z] \cdot \mathbb{E}[X]$

So uncorrelated.

But $\mathbb{P}(X = 2, Z = 1) \neq \mathbb{P}(X = 2)\, \mathbb{P}(Z = 1)$

$\downarrow$ 0 $\quad$ $\downarrow \neq 0$ $\quad$ $\downarrow \neq 0$

So X, Z are not independent.

Uncorrelatedness $\Longrightarrow$ Independence only when

① Jointly Gaussian RVs.

② Binary RVs.

We will see ① later. Take ②

$X \sim Bn(p)$
$Y \sim Bn(q)$

Say $\mathbb{E}[XY] = \mathbb{E}[X] \cdot \mathbb{E}[Y]$

$\mathbb{P}(X=1, Y=1) \quad \mathbb{P}(X=1) \quad \mathbb{P}(Y=1)$

Now check for other tuples

$$\begin{aligned}\mathbb{P}(X=1, Y=0) &= \mathbb{P}(X=1) - \mathbb{P}(X=1, Y=1) \\ &= \mathbb{P}(X=1) - \mathbb{P}(X=1)\,\mathbb{P}(Y=1) \\ &= \mathbb{P}(X=1)\,(1 - \mathbb{P}(Y=1)) \\ &= \mathbb{P}(X=1)\,P(Y=0)\end{aligned}$$

Similarly others.

So $X \perp\!\!\!\perp Y$.